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Project Proposal

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Security Audit Tool: A Modular C++ Based Security Assessment System

1. Introduction

In today's digital era, cybersecurity has become a critical concern for organizations and individuals. Systems connected to the internet are constantly exposed to security threats such as unauthorized access, open ports, misconfigured services, and information leakage. Security auditing helps in identifying such vulnerabilities before they can be exploited.

The proposed project, **Security Audit Tool**, is a modular command-line based application developed in C++ that performs essential security assessment and reconnaissance tasks. The system is designed to demonstrate the practical implementation of Object-Oriented Programming (OOP) concepts while providing real-world cybersecurity functionality.

2. Problem Statement

Many professional security auditing tools available in the market are complex, resource-intensive, and difficult for students to understand from a programming perspective. There is a need for a simplified yet technically structured tool that demonstrates how security auditing mechanisms work internally.

This project aims to design and implement a lightweight, database-free security audit tool using C++, focusing on modular design and core security functionalities.

3. Objectives

The main objectives of this project are:

a) To design and develop a modular Security Audit Tool using C++.
b) To apply Object-Oriented Programming concepts such as abstraction, inheritance, and polymorphism.
c) To implement WHOIS lookup functionality for domain information gathering.
d) To develop a basic TCP port scanning module.
e) To perform DNS lookup for domain resolution details.
f) To generate structured security audit reports in text format.
g) To create a professional ASCII-based terminal interface.

4. Scope of the Project

The Security Audit Tool will include the following core modules:

4.1 WHOIS Lookup Module

a) Retrieves domain registration information.
b) Displays registrar, creation date, and expiration date.

4.2 Port Scanner Module

a) Scans commonly used TCP ports (e.g., 21, 22, 80, 443).
b) Identifies open and closed ports.

4.3 DNS Lookup Module

a) Retrieves A records and Name Server (NS) details.
b) Displays domain resolution information.

4.4 Report Generation Module

a) Saves scan results in a structured text file.
b) Maintains session-based logs in memory using C++ STL containers.

The system will not use a database in order to keep the architecture lightweight and portable.

5. Methodology

The development of the Security Audit Tool will follow a modular and object-oriented approach:

a) Design an abstract base class (e.g., Scanner) to define common scanning behavior.
b) Implement derived classes for WHOIS, Port Scanning, and DNS Lookup modules.
c) Use C++ Standard Template Library (STL) structures such as vector for in-memory data handling.
d) Implement socket programming for port scanning.
e) Develop a structured command-line interface with ASCII formatting.
f) Implement file handling mechanisms to generate audit reports.

6. Tools and Technologies

a) **Programming Language:** C++
b) **Programming Paradigm:** Object-Oriented Programming (OOP)
c) **Libraries:** Standard C++ Libraries, Socket Programming
d) **Platform:** Linux / Windows
e) **Compiler:** GCC / MinGW

7. Expected Outcomes

Upon successful completion, the project will:

a) Demonstrate practical implementation of OOP and Data Structures.
b) Provide a functional command-line based security auditing tool.
c) Generate readable and structured security reports.
d) Serve as a foundation for future expansion into advanced cybersecurity tools.

8. Future Enhancements

Potential future improvements include:

a) Integration of a database for persistent storage.
b) Graphical User Interface (GUI) implementation.
c) Advanced port scanning techniques.
d) Automated vulnerability detection.
e) Integration with external security APIs.

9. Conclusion

The Security Audit Tool is designed to combine academic programming knowledge with practical cybersecurity applications. By implementing modular architecture and core security functionalities in C++, the project provides both educational value and real-world relevance. It establishes a strong foundation for further development in the field of cybersecurity and software engineering.